

TÍCH PHÂN EULER

(Sử dụng hàm Gamma và Beta để tính các tích phân sau)

1. $A = \int_0^{+\infty} x^{2n} e^{-x^2} dx$

Hướng dẫn giải

+) Ta có: $A = \int_0^{+\infty} (x^2)^n e^{-x^2} dx = \frac{1}{2} \int_0^{+\infty} (x^2)^{n-\frac{1}{2}} e^{-x^2} d(x^2) = \frac{1}{2} \int_0^{+\infty} (x^2)^{(n+\frac{1}{2})-1} e^{-x^2} d(x^2)$

+) Đặt $x^2 = t$; ta có tại $x = 0$ thì $t = 0$, khi $x \rightarrow +\infty$ thì $t \rightarrow +\infty$

+) Khi đó:

$$A = \frac{1}{2} \int_0^{+\infty} t^{(n+\frac{1}{2})-1} e^{-t} dt = \frac{1}{2} \cdot \Gamma\left(n + \frac{1}{2}\right) = \frac{1}{2} \cdot \frac{(2n)!}{n! \cdot 2^{2n}} \cdot \sqrt{\pi} = \frac{(2n)! \cdot \sqrt{\pi}}{n! \cdot 2^{2n+1}}.$$

2. $B = \int_0^2 x^6 \sqrt[3]{(8-x^3)^2} dx.$

Hướng Dẫn Giải

+) Ta có: $B = \int_0^2 x^6 \sqrt[3]{(8-x^3)^2} dx = 4 \int_0^2 x^6 \sqrt[3]{\left[1 - \left(\frac{x}{2}\right)^3\right]^2} dx$

$$= 4 \int_0^2 x^6 \left[1 - \left(\frac{x}{2}\right)^3\right]^{\frac{2}{3}} dx.$$

+) Đặt $\left(\frac{x}{2}\right)^3 = t \Rightarrow \begin{cases} x = 2\sqrt[3]{t} \\ dx = \frac{2}{3} t^{-\frac{2}{3}} dt \end{cases}$

Đổi cận:

x	0	2
t	0	1

+) Khi đó :

$$\begin{aligned} B &= 4 \int_0^1 (2\sqrt[3]{t})^6 (1-t)^{\frac{2}{3}} \cdot \frac{2}{3} t^{-\frac{2}{3}} dt = 4 \cdot 2^6 \cdot \frac{2}{3} \int_0^1 t^{\frac{4}{3}} (1-t)^{\frac{2}{3}} dt \\ &= \frac{2^9}{3} B\left(\frac{7}{3}, \frac{5}{3}\right) = \frac{2^9}{3} \frac{\Gamma\left(\frac{7}{3}\right) \Gamma\left(\frac{5}{3}\right)}{\Gamma(4)} \\ &= \frac{2^9}{3} \frac{4}{3} \frac{1}{3} \frac{2}{3} \frac{\Gamma\left(\frac{1}{3}\right) \Gamma\left(\frac{2}{3}\right)}{3!} \quad \left(\text{do } \Gamma(x+1) = x \cdot \Gamma(x) \right) \\ &= \frac{2^{11}}{3^5} \frac{\pi}{\sin \frac{\pi}{3}} = \frac{2^{12} \pi}{3^5 \sqrt{3}}. \end{aligned}$$

$$3. C = \int_0^1 \frac{dx}{\sqrt[n]{1-x^n}}$$

Hướng dẫn giải

$$+) \text{ Đặt } x^n = t \Rightarrow \begin{cases} nx^{n-1}dx = dt \\ dx = \frac{1}{n} \cdot t^{\frac{1}{n}-1} dt \end{cases}$$

$$\text{Đổi cận: } \begin{array}{c|c|c} x & 0 & 1 \\ \hline t & 0 & 1 \end{array}$$

+) Khi đó ta có:

$$C = \int_0^1 \frac{\frac{1}{n} \cdot t^{\frac{1}{n}-1} dt}{(1-t)^{\frac{1}{n}}} = \frac{1}{n} \int_0^1 t^{\frac{1}{n}-1} (1-t)^{-\frac{1}{n}} dt = \frac{1}{n} B\left(\frac{1}{n}, 1 - \frac{1}{n}\right) = \frac{1}{n} \cdot \frac{\pi}{\sin \frac{\pi}{n}}$$

$$4. D = \int_0^{\frac{\pi}{2}} \sin^4 \varphi \cos^6 \varphi d\varphi$$

Hướng dẫn giải

$$+) \text{ Ta có: } D = \int_0^{\frac{\pi}{2}} \sin^4 \varphi \cos^6 \varphi d\varphi = \frac{1}{2} B\left(\frac{5}{2}, \frac{7}{2}\right) = \frac{1}{2} \frac{\Gamma\left(\frac{5}{2}\right) \Gamma\left(\frac{7}{2}\right)}{\Gamma(6)}$$

$$+) \text{ Mà } \begin{cases} \Gamma\left(\frac{7}{2}\right) = \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \cdot \Gamma\left(\frac{1}{2}\right) = \frac{15\sqrt{\pi}}{8} \\ \Gamma\left(\frac{5}{2}\right) = \frac{3}{2} \cdot \frac{1}{2} \cdot \Gamma\left(\frac{1}{2}\right) = \frac{3\sqrt{\pi}}{4} \\ \Gamma(6) = 5! = 120 \end{cases}$$

$$\text{Vậy } D = \frac{1}{2} \cdot \frac{15\sqrt{\pi}}{8} \cdot \frac{3\sqrt{\pi}}{4} \cdot \frac{1}{120} = \frac{3\pi}{512}$$

$$5. E = \int_0^{\pi} \sin^{99} \varphi d\varphi$$

Hướng dẫn giải

$$+) \text{ Đặt } t = \frac{\pi}{2} - \varphi \Rightarrow \begin{cases} \varphi = \frac{\pi}{2} - t \\ dt = -d\varphi \end{cases}$$

$$\text{Đổi cận: } \begin{array}{c|c|c} \varphi & 0 & \pi \\ \hline t & \frac{\pi}{2} & -\frac{\pi}{2} \end{array}$$

Suy ra: $E = \int_0^{\pi} \sin^{99} \varphi d\varphi = \int_{\frac{\pi}{2}}^{-\frac{\pi}{2}} -\sin^{99} \left(\frac{\pi}{2} - t \right) dt = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^{99} t dt$

+) Do $\cos^{99} t$ là hàm chẵn trên $\left[-\frac{\pi}{2}; \frac{\pi}{2}\right]$ nên ta có:

$$E = 2 \int_0^{\frac{\pi}{2}} \cos^{99} t dt = 2 \cdot \frac{1}{2} B\left(\frac{1}{2}, 50\right) = B\left(\frac{1}{2}, 50\right)$$

+) Mà $B\left(\frac{1}{2}, 50\right) = \frac{49}{49 + \frac{1}{2}} B\left(\frac{1}{2}, 49\right) = \dots = \frac{49}{\frac{99}{2}} \cdot \frac{48}{\frac{97}{2}} \cdot \dots \cdot \frac{1}{\frac{3}{2}} \cdot B\left(\frac{1}{2}, 1\right) = \frac{49!}{\frac{1}{2} \cdot \frac{3}{2} \cdot \dots \cdot \frac{99}{2}} = \frac{2^{50} 49!}{99!!}$

Vậy $E = \frac{2^{50} 49!}{99!!}$

6. $F = \int_0^{\frac{\pi}{4}} \sin^{100} \varphi \sin 4\varphi d\varphi$

Hướng dẫn giải

+) Ta có: $F = \int_0^{\frac{\pi}{4}} \sin^{100} \varphi \sin 4\varphi d\varphi = \int_0^{\frac{\pi}{4}} \left(\frac{1 - \cos 2\varphi}{2} \right)^{50} 2 \cos 2\varphi \sin 2\varphi d\varphi$

+) Đặt $t = \cos 2\varphi \Rightarrow dt = -2 \sin 2\varphi d\varphi$

Đổi cận: $\begin{array}{c|c|c} \varphi & 0 & \frac{\pi}{4} \\ \hline t & 1 & 0 \end{array}$

+) Khi đó: $F = \int_0^1 \left(\frac{1-t}{2} \right)^{50} t dt = \frac{1}{2^{50}} \int_0^1 t(1-t)^{50} dt = \frac{1}{2^{50}} B(2, 51)$
 $= \frac{1}{2^{50}} \cdot \frac{\Gamma(2)\Gamma(51)}{\Gamma(53)} = \frac{1}{2^{50}} \cdot \frac{1! \cdot 50!}{52!} = \frac{1}{2^{52} \cdot 663}$

7. $G = \int_0^{+\infty} \frac{\sqrt[4]{x}}{(1+x^2)^8} dx$

Hướng dẫn giải

+) Đặt $t = x^2 \Rightarrow \begin{cases} x = \sqrt{t} \\ dx = \frac{dt}{2\sqrt{t}} \end{cases}$ Tại $x = 0$ thì $t = 0$, khi $x \rightarrow +\infty$ thì $t \rightarrow +\infty$

Suy ra: $G = \frac{1}{2} \int_0^{+\infty} \frac{t^{-\frac{3}{8}}}{(1+t)^8} dt = \frac{1}{2} B\left(\frac{5}{8}, \frac{59}{8}\right)$

+) Có: $B\left(\frac{5}{8}, \frac{59}{8}\right) = \frac{\Gamma\left(\frac{5}{8}\right)\Gamma\left(\frac{59}{8}\right)}{\Gamma(8)}$

Mà $\begin{cases} \Gamma\left(\frac{59}{8}\right) = \frac{51}{8} \cdot \frac{43}{8} \cdot \frac{35}{8} \cdot \frac{27}{8} \cdot \frac{19}{8} \cdot \frac{11}{8} \cdot \frac{3}{8} \cdot \Gamma\left(\frac{3}{8}\right) \\ \Gamma\left(\frac{3}{8}\right)\Gamma\left(\frac{5}{8}\right) = \frac{\pi}{\sin \frac{3\pi}{8}} \\ \Gamma(8) = 7! \end{cases}$

Vậy: $G = \frac{1}{2}B\left(\frac{5}{8}, \frac{59}{8}\right) = \frac{51 \cdot 43 \cdot 35 \cdot 27 \cdot 19 \cdot 11 \cdot 3 \cdot \pi}{2^{22} \cdot 7! \cdot \sin \frac{3\pi}{8}}$

8. $H = \int_{-1}^1 \frac{(1-x)^{2p-1}(1+x)^{2q-1}}{(1+x^2)^{p+q}} dx$

Hướng dẫn giải

+) Đặt $x = \tan\left(\varphi - \frac{\pi}{4}\right) \Rightarrow dx = \frac{2(\tan^2 \varphi + 1)}{(\tan \varphi + 1)^2} d\varphi$

Đổi cận: $\begin{array}{c|c|c} x & 1 & -1 \\ \hline \varphi & \frac{\pi}{2} & 0 \end{array}$

+) Khi đó: $H = \int_0^{\frac{\pi}{2}} \frac{\left(1 - \frac{\tan \varphi - 1}{\tan \varphi + 1}\right)^{2p-1} \left(1 + \frac{\tan \varphi - 1}{\tan \varphi + 1}\right)^{2q-1}}{\left[1 + \left(\frac{\tan \varphi - 1}{\tan \varphi + 1}\right)^2\right]^{p+q}} \cdot \frac{2(\tan^2 \varphi + 1)}{(\tan \varphi + 1)^2} d\varphi$

$$= \int_0^{\frac{\pi}{2}} \frac{\left(\frac{2}{\tan \varphi + 1}\right)^{2p-1} \left(\frac{2 \tan \varphi}{\tan \varphi + 1}\right)^{2q-1}}{\left[\frac{2(\tan^2 \varphi + 1)}{(\tan \varphi + 1)^2}\right]^{p+q-1}} d\varphi$$

$$= \int_0^{\frac{\pi}{2}} \frac{2^{2p+2q-2} \cdot \tan \varphi^{2q-1} \cdot (\tan \varphi + 1)^{2p+2q-2}}{(\tan \varphi + 1)^{2p+2q-2} \cdot 2^{p+q-1} \cdot (\tan^2 \varphi + 1)^{p+q-1}} d\varphi$$

$$= 2^{p+q-1} \int_0^{\frac{\pi}{2}} \tan \varphi^{2q-1} \cos \varphi^{2p+2q-2} d\varphi$$

$$= 2^{p+q-1} \int_0^{\frac{\pi}{2}} \sin \varphi^{2q-1} \cos \varphi^{2p-1} d\varphi$$

$$= 2^{p+q-2} B(p, q)$$